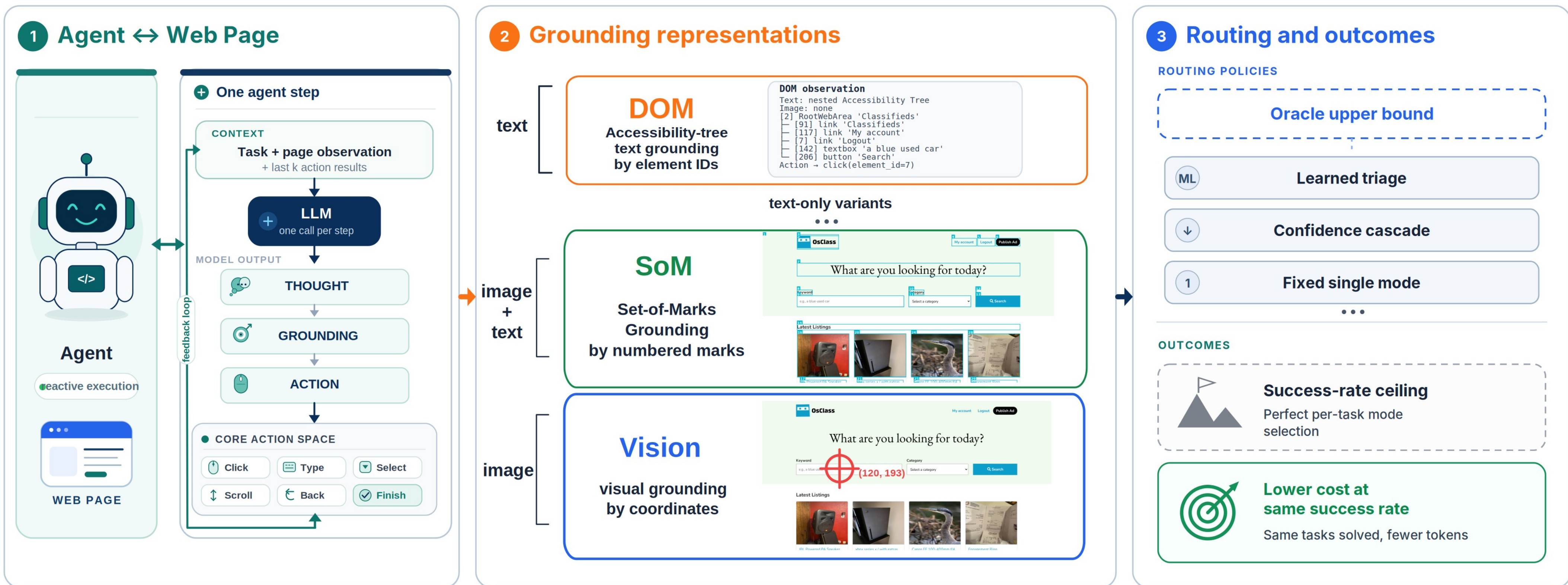


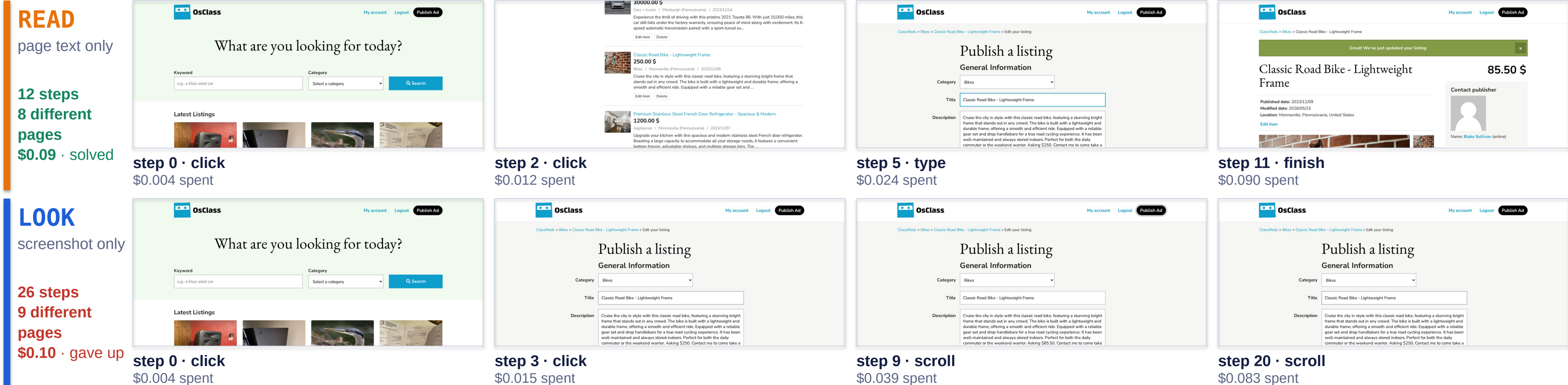
Measuring the ceiling, the predictability and the economics of representation routing in web agents

WHAT AN AGENT SEES, AND WHO CHOOSES

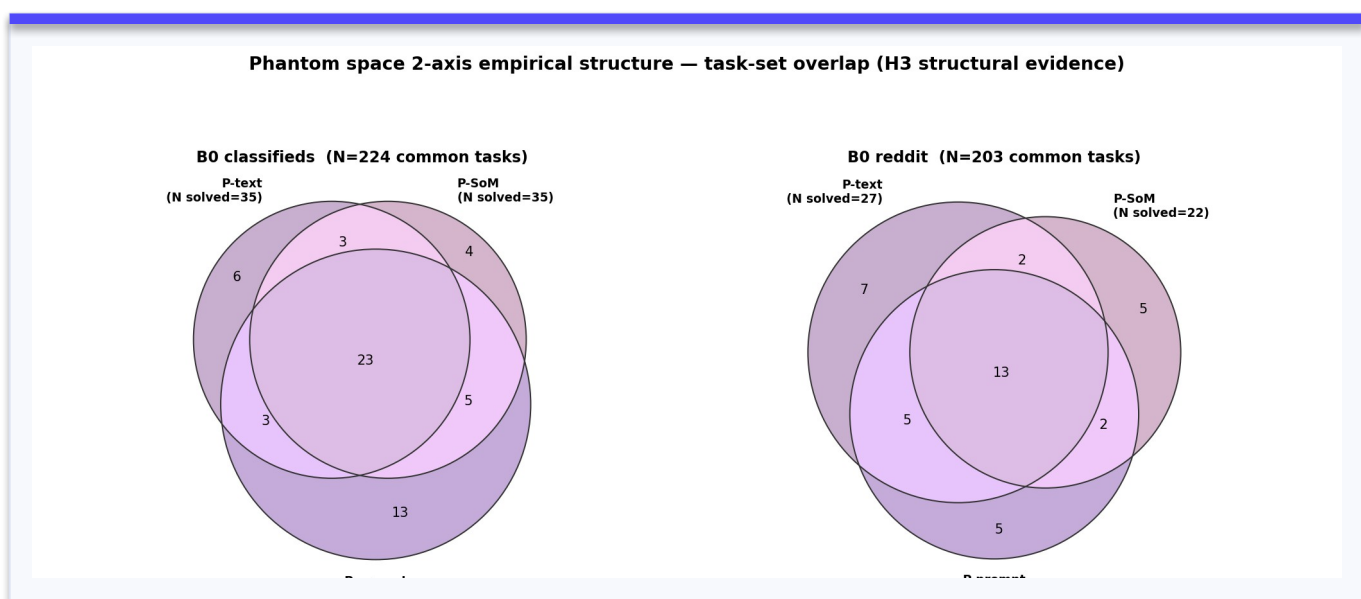


THE SAME TASK, SEEN TWO WAYS

“Go to my listing of the blue bike and change the price to \$85.50 — and say so in the description.”



DIFFERENT VIEWS, DIFFERENT TASKS



Tasks solved by each text-only view. The sets overlap; they do not coincide.

THEY ALSO BEHAVE DIFFERENTLY

Across 26 behavioural measures, how often a view is the extreme one in at least 7 of the 8 settings:

view	measures
Vision — screenshot only	9
SoM — screenshot + text	5
DOM — page text only	0
the 3 other text-only views	0

The extremes sit with the two views that see the page. How the agent moves is set by whether it gets a picture, not by which text it gets.

SIX VIEWS, EIGHT SETTINGS

	VisualWebArena				WebArena			
DOM	17.4	14.3	6.2	5.9	1.3	3.9	26.9	16.4
P-prompt	19.6	12.3	6.7	5.4	1.8	0.0	26.0	16.4
P-text	15.6	13.3	7.6	5.9	0.5	2.0	35.6	16.4
P-SoM	15.6	10.8	6.7	5.9	0.9	0.5	25.0	11.5
SoM	27.2	14.8	14.3	7.4	2.2	1.0	22.1	13.5
Vision	25.0	7.4	12.5	2.5	2.2	2.0	19.2	9.6

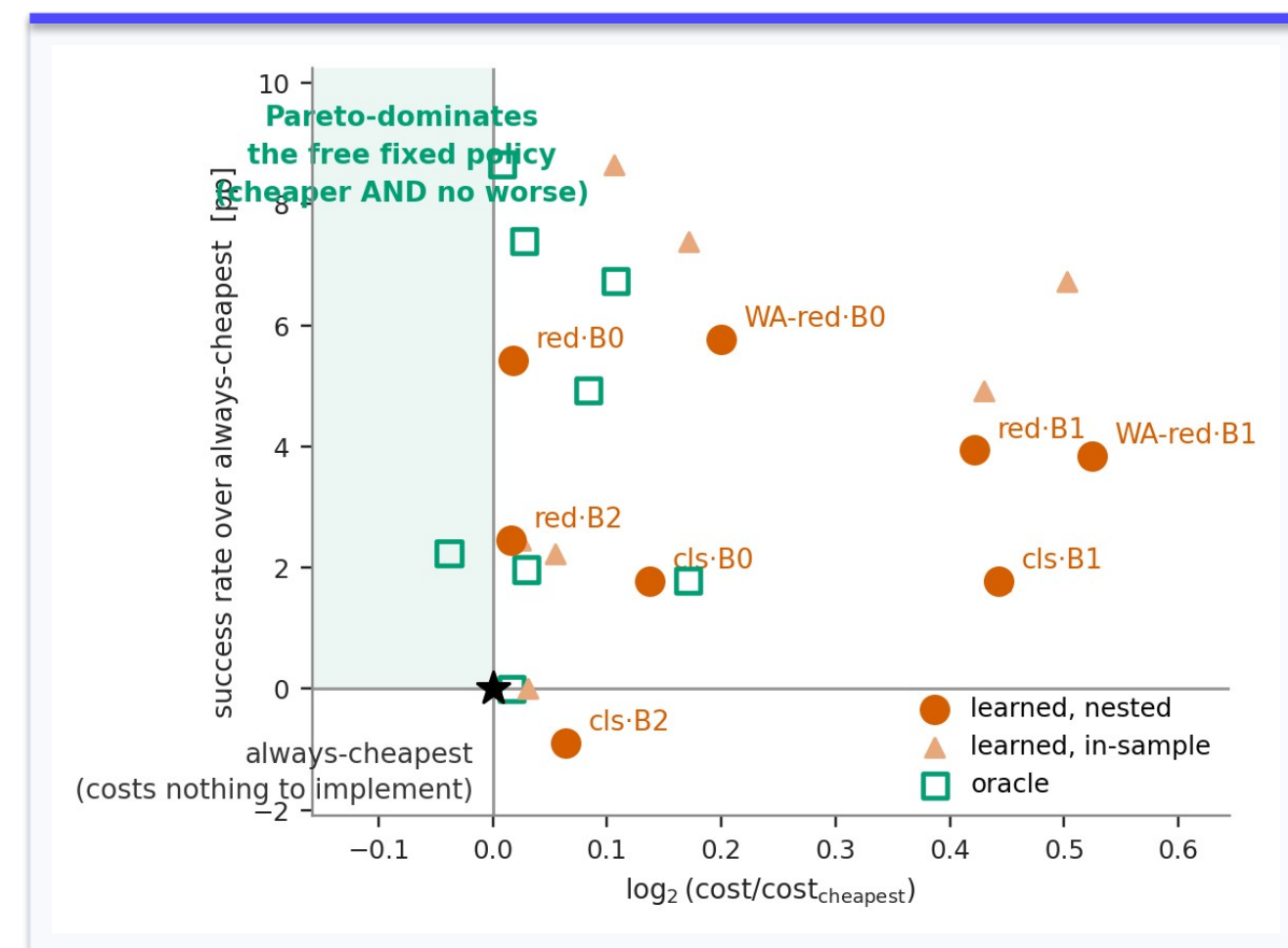
Success rate (%). Colour is what reaches the model: text, text + image, image.

AND THEY FAIL DIFFERENTLY

On tasks only one side solved, how did the loser fail? × = versus its own everyday failure rate.

When the picture won, the text-only agent gave up once it could not find it	2.3×
could not see what the task asked about	2.2×
When the text won, the picture-only agent never turned the page	0.9×
ran out of budget, unfinished	0.9×
One side dies of something you can name. The other simply never arrives.	

CHOOSING WELL IS HARDER THAN IT LOOKS



Each policy against always using the cheapest view. A win lands in the shaded region — none does.

And the target is small: running one unchanged view twice already flips 10–14% of tasks.

SO WHAT

The views are worth choosing between — they solve different tasks, move differently and fail differently.

Choosing well, in advance, is the part that does not work yet.

